

RAVIKULA SILVA

ELECTRICAL GRADUATE ENGINEER - BUILDING SERVICES

Melbourne, VIC · +61 431 638 202 · sravikula@gmail.com · [LinkedIn](#)

PROFILE SUMMARY

Electrical Building Services Engineer with hands-on experience in the design, estimation, and documentation of electrical building services systems across residential and commercial projects, from concept through to commissioning. Proficient in lighting design (AS 1680, DIALux, DALI, KNX/ETS), electrical load calculations, cable sizing, and protection coordination in compliance with AS 3000, AS 3008, and the NCC. Experienced in producing and updating electrical schematics, SLDs, and shop drawings in AutoCAD and Revit, and validating designs using PowerCAD. Proactive and organised team player with a proven ability to coordinate across building services project teams in Australia.

TECHNICAL & CORE COMPETENCIES

Electrical Design & Documentation: AutoCAD, Revit (foundational), PowerCAD, DIALux, Electrical Schematics (SLD) Interpretation, Cable sizing and Protection Coordination

Automation: KNX, ETS, DALI Control, Zencontrol, PLC Programming (Siemens), Python

Standards & Compliance: Australian Standards (AS 3000, AS 3008, AS 1680), NCC

Project Delivery: Project Coordination & Estimation, Site Commissioning, Technical Documentation, Bluebeam Markups

WORK EXPERIENCE

iAutomation Pty Ltd

Intern Engineer

Victoria, Australia

Jan 2026 – Apr 2026

- Supported design and estimation of electrical building services and lighting control systems across residential and commercial projects, from concept through to commissioning, contributing to successful delivery of a healthcare facility project valued at AUD 120K.
- Maintained electrical schematics, single-line diagrams, and shop drawings in AutoCAD across active project phases, ensuring documentation remained accurate against live site conditions and supporting efficient handover to contractors.
- Attended site commissioning and troubleshooting activities alongside senior engineers, supporting system verification and on-site issue resolution to keep projects on track.
- Programmed KNX-based control systems using ETS across multiple building automation projects, delivering scene control, DALI dimming, relay switching, and BMS interfacing to meet client lighting and automation requirements.
- Coordinated with construction contractors and electrical wholesalers throughout project delivery and commissioning, supporting smooth procurement and on-site execution of building automation systems.

Rom-Control Pty Ltd

Electronics Engineering Intern

Victoria, Australia

Sep 2025 – Dec 2025

- Tested and validated electrical control systems including VSDs and power supply units using schematics and test equipment, developing hands-on diagnostic skills directly applicable to building services commissioning.
- Assessed semiconductor devices for functional reliability, building a systematic approach to fault identification and component verification applicable to protection and switching systems.

Synopsys Inc.

Application Engineer

Mountain View, CA (Remote)

Jul 2022 – Jul 2023

- Automated repetitive engineering workflows using Python scripting, reducing manual effort and improving process consistency across validation tasks.
- Conducted systematic product validation and debugging, producing structured technical documentation that supported clearer identification and resolution of design issues.

South Asia Gateway Terminals Pvt Ltd

Electrical Engineering Intern

Sri Lanka

Oct 2020 – Apr 2021

- Designed and implemented a capacitor bank system that improved site power factor from 0.85 to 0.95 and reduced reactive power consumption by 12%, demonstrating applied power quality design within a live operational electrical installation.
- Developed PLC-based automated control systems using Siemens S7-300, reducing manual crane operation by 90% and labour costs by 30%, gaining hands-on experience in electrical control design, commissioning, and system integration.

RELEVANT PROJECTS

Electrical Building Services Design, Residential and Commercial (Point Cook, VIC) <i>RMIT University, Australia</i>	Mar 2025 – Jun 2025
- Delivered a complete electrical building services design for a residential dwelling and a detached commercial graphic arts studio, covering HVAC, lighting, and full electrical installation across both project types, demonstrating multi-building services design capability. - Designed the submains electrical supply connecting the residential distribution board to a detached commercial studio 72m away, applying cable sizing, voltage drop verification, and protection coordination across both distribution boards to meet AS 3000 compliance. - Designed indoor and outdoor lighting systems for residential and commercial spaces in compliance with AS 1680, using DIALux to calculate illuminance levels, room indices, and fitting layouts, producing compliant lighting plans for all functional areas. - Performed maximum demand calculations, sub-circuit design, cable sizing, and protection coordination in accordance with AS 3000 and AS 3008.1, validating the full design in PowerCAD and achieving an overall voltage drop of 1.25% against a 3.35% permissible limit. - Produced single-line diagrams (SLDs) and complete electrical documentation including protection device schedules, cable specifications, and circuit layouts for residential and commercial distribution boards, delivering a fully documented design package.	
Residential Microgrid Design, Fairfield VIC <i>RMIT University, Australia</i>	Jul 2024 – Sep 2024
- Designed a residential hybrid microgrid integrating a 2.5 kW solar PV array, 2 kW inverter, and 15 kWh battery storage, applying load analysis and system sizing principles applicable to building electrical design. - Performed techno-economic analysis across four dispatch strategies using HOMER Pro, evaluating NPC, LCOE, and fuel consumption to identify the most cost-effective system configuration.	
Automated Small Scale Manufacturing System Design <i>RMIT University, Australia</i>	Jul 2023 - Sep 2023
- Designed and implemented PLC-based automation using Siemens TIA Portal for a modular manufacturing line integrating motor control, pneumatic systems, and sensor-based operations, delivering a fully functional automated system. - Diagnosed PLC and actuator faults and supported commissioning of control systems, applying structured fault-finding methods to ensure reliable system operation and optimised equipment performance.	

EDUCATION

RMIT University <i>MSc in Electronic and Electrical Engineering GPA: 3.6/4 (Distinction)</i>	Melbourne, Australia Jul 2025
University of Moratuwa <i>BSc (Hons) in Electrical Engineering GPA: 3.7/4.2 (Second Class Upper Division, equivalent to Distinction)</i>	Sri Lanka Jun 2022

PROFESSIONAL QUALIFICATIONS AND CERTIFICATIONS

National Electricity Market (NEM) Foundations AEMO eLearning	Apr 2026
Construction White Card, Australia	Mar 2026
Graduate Member, Engineers Australia (GradIEAust)	Since July 2025

LEADERSHIP & VOLUNTEERING

Rotaract Club of University of Moratuwa <i>Vice President – Professional Development</i>	Sri Lanka Jul 2021 – Jun 2022
- Directed and delivered the university’s officially recognised career fair across four faculties (Engineering, Architecture, IT, Business), engaging 600+ undergraduates and 80+ companies. - Led a team of 150+ volunteers across planning, stakeholder coordination, and execution, resulting in 200+ students progressing to next stages in recruitment processes.	
Media Lead – Project Zooxanthellae Coral Reef Restoration Initiative	Jul 2018 – Jun 2019
Led media and communications for the final year of a three-year coral reef restoration project, managing content, social media, and stakeholder engagement, while supporting project visibility and impact reporting, contributing to the project receiving a Rotaract Outstanding Project Award (South Asia).	

REFERENCES

Jason Wong Head of Lighting Control – iAutomation Pty Ltd +61 458 300 739 jasonw@iautomation.com.au
Don Roshan Project and Systems Engineer – iAutomation Pty Ltd +61 407 133 019 dons@iautomation.com.au